Innovation management

How might we solve the problem of social isolation around sharing and sustainability ?

Poppins' sharing boxes

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I. Introduction

Have you ever found yourself in need of an object that you only need for a short period of time or a specific activity? Perhaps a paddle, a racletière or a volleyball. Buying and keeping these items can be costly and impractical, especially for students who often are on a tight budget and have small apartments.

That’s where our Box Sharing concept comes in. With this innovative idea, students can put objects that they own but don't use regularly in a specific locked box, and others can borrow them for a short period of time. With this concept comes also an app (see Figure A3 in Appendix), where you can see which objects are available and allow you to book them and unlock the box with the desired object. When you retrieve an object you have to rate how the previous user left the object, this to ensure good maintenance of the object. This not only helps students save money and space but also promotes a sense of community and sharing among students.

Overall, the Box Sharing concept is a practical and cost-effective and eco-friendly solution for students who want to share resources and help each other succeed.

II. Pre-business plan

The Swiss economy is known for being stable and innovative, and the country has a high standard of living. There is also a growing trend towards sustainable and collaborative living, which could make our sharing platform an attractive option for many people (see survey on Figure B1 in Appendix).
In terms of competition, there are a few existing sharing platforms in Switzerland, such as Sharely and Sharoo. However, Poppins’ unique selling point could be the focus on lending and borrowing physical objects, rather than just services or experiences.

The 268'000 students in Switzerland in 2021 (constantly rising number) have around 27'600 places available in student residences. We have chosen “the Vortex” as our early adopters. The vortex is the largest student residence of Switzerland of a single owner. This owner is the Fondation Maison pour Etudiants Lausanne (FMEL). The place hosts around 1000 students and its proximity to EPFL and UNIL campus, and the subway M1 makes it very popular. This is where we will put our boxes first.
Then, we plan to extend to other residences of FMEL around Lausanne, and similar places like Les Estudiantines or Stud’home. We plan to sell each box for a price of 500 CHF. This will allow us to cover our marketing and fabrication costs. To see if this option seems viable, we talked to Mrs Garrido, administrative manager at FMEL, who seems really interested in our project. Indeed, our innovation helps to create a virtuous circle and promote values in common with FMEL, like contact between people, friendship and solidarity.

III. Technical concerns

The technical development work is divided into two parts, the hardware and the software. The electrical diagram (Fig. A1) and the list of components (Fig. A2) are shown in the Appendix.

Some critical design choices have been made, they are listed below.

- Plexiglass windows: Plexiglas windows allow users to see the objects contained in the lockers. This material is a strong thermoplastic polymer known for its impact resistance and light weight.
- Arduino Nano: The Arduino Nano 33 IoT is a reliable and low power consuming micro-controller. With its integrated Bluetooth module, it allows an easy and compact integration.
- Bluetooth phone app: The Bluetooth communication between the smartphone and the locker allows a great accessibility of the service. In fact, the application will emit a secret code specific to the locker that will open automatically. The code does not need to be renewed at each use as no user has access to it properly.
- Several sizes of boxes: The size of the boxes is adapted to the size of the object in order to take up as little space as possible in the hall of the residence.
- Power supply: the module has a cable that links it to the mains. This choice was made to the detriment of a battery for practical reasons (no need to recharge it manually...) and for sustainability reasons. Also the module is not intended to be placed in outdoor and mobile locations so a mains power supply is very suitable.

IV. Management

In our project, the management component is crucial for ensuring the team's success. We employ a structured and participatory approach, emphasizing objective setting, collaborative decision-making, and use of SMART goals. By implementing these strategies, we aim to enhance team productivity, accountability, and effective communication, ultimately leading to the successful execution of our project.

Our business plan embraces an Agile Strategy, allowing us to quickly adapt to market changes and seize new opportunities. This approach offers several key advantages:

- Flexibility: It is possible to quickly adjust the features of the sharing boxes according to the changing needs of the residents. For example, if residents express the need to add new types of items to share, the team can rapidly integrate this functionality.
- Collaboration: The agile approach promotes collaboration among residents, building managers, and other stakeholders. A stakeholder analysis is shown in Figure B2 in Appendix. For instance, residents can suggest improvements or ideas for the sharing boxes, fostering a collaborative environment for innovation.
- Continuous value delivery: The agile approach enables the regular delivery of usable features. This means that as soon as the first sharing boxes are ready, residents can

start using them, even if not all planned features have been developed yet. This allows for early value realization.

- Adaptability: The agile approach allows for adaptation to market trends or resident demands. For example, if more sharing boxes are needed in a building, discussions can be held with the facility management to accommodate the demand. This ensures that the sharing boxes are better suited to residents' actual needs, leading to a positive user experience and increased usage.

A Gantt Chart is used (see Figure C1 in Appendix) to visually represent project timelines, tasks, and dependencies, providing a clear overview of the project's schedule.
A Stakeholder Analysis in Figure B2 (Appendix) shows the different partners and organizations that Poppins has to work with for launching the product. It gives a good overview of all the interactions between Poppins, its clients and its suppliers.

Regarding payment methods, the options mentioned include collecting credit card information to determine if a user should be banned, billing through FMEL or credit cards, or setting a minimum price on the application to ensure billing capability in case of any issues. If the borrowed item is returned in good condition, 0.- are charged to the borrower. If a borrower notices that the borrowed item is in bad condition, he/she can report it on the application and a delegate is in charge of settling the dispute. The delegate may decide to charge the borrower for the repair of the damaged item.

V. Marketing

The name Poppins has been chosen because of the film and book ‘Mary Poppins’. In the story, the eponymous character has a bag in which a large range of objects are stored. One of these objects is an umbrella and has been chosen to be the brand’s logo.

The communication is based on a diffusion on Instagram and Tiktok, in order to reach the young EPFL’s and FMEL’s community.

Appendix

A. Technical concerns

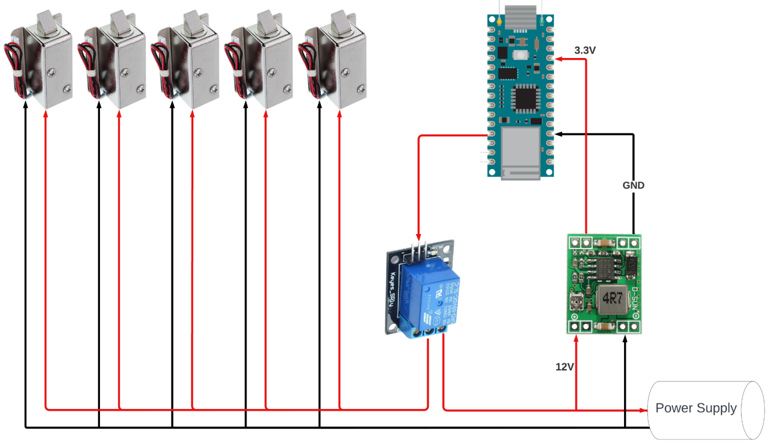
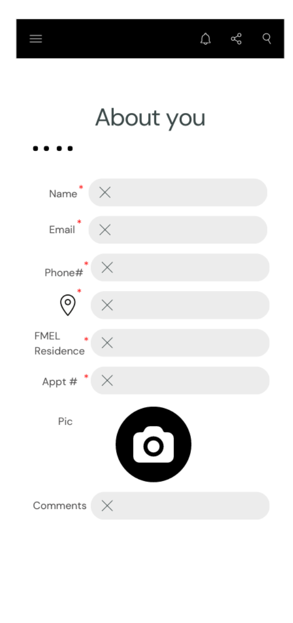
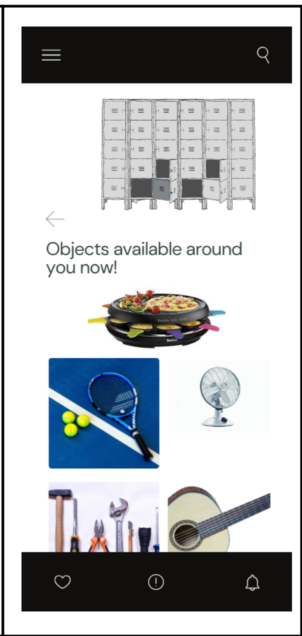


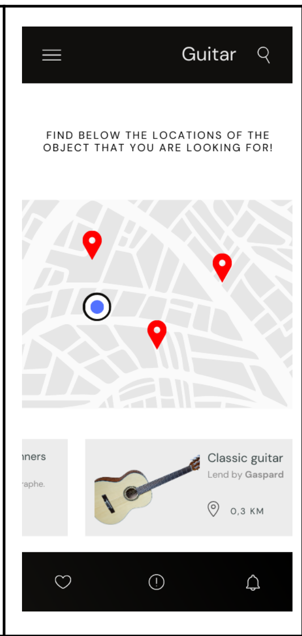
Figure A1 - Electrical schematic of 5 smart locks

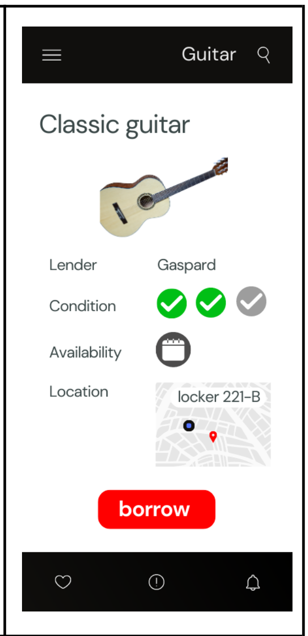
Item	Price (in CHF)
Arduino Nano 33 IoT	21
Mini Electromagnetic Solenoid Lock Safety Cabinet Drawer Lock	5 x 4
DC-DC convertor 12V to 3.3V	2.70
Relay for lock	2.40
12V power supply	6.35
See through plastic board	7.75
Wood board	20
Miscellaneous screws and wires	5
Total	**85.2**

Figure A2 - List of compounds and prices for 5 lockers





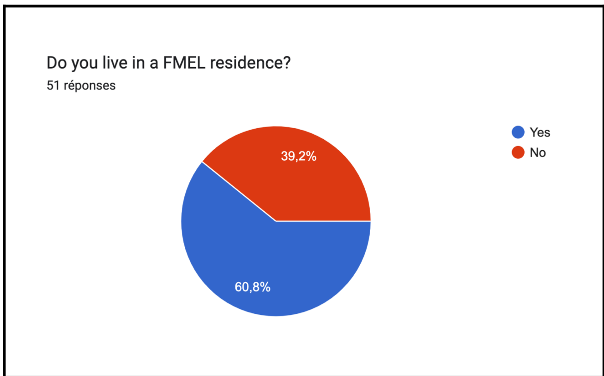


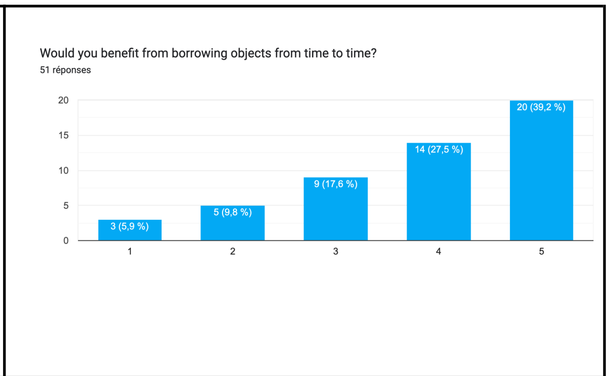


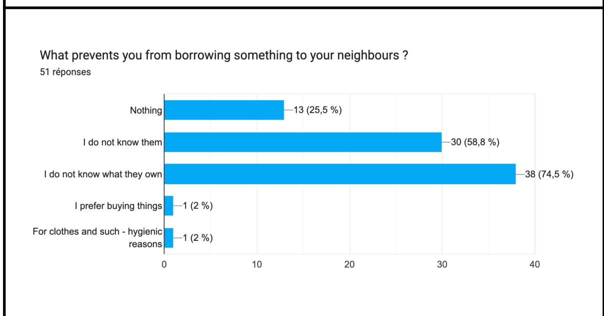
Registration form	List of available items around	“Around you” map	Item borrowing page

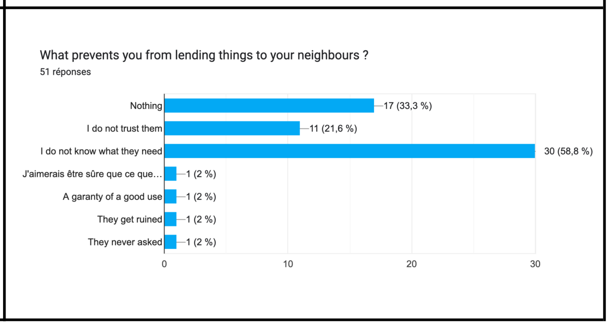
Figure A3 - App screenshots

B. Management and marketing









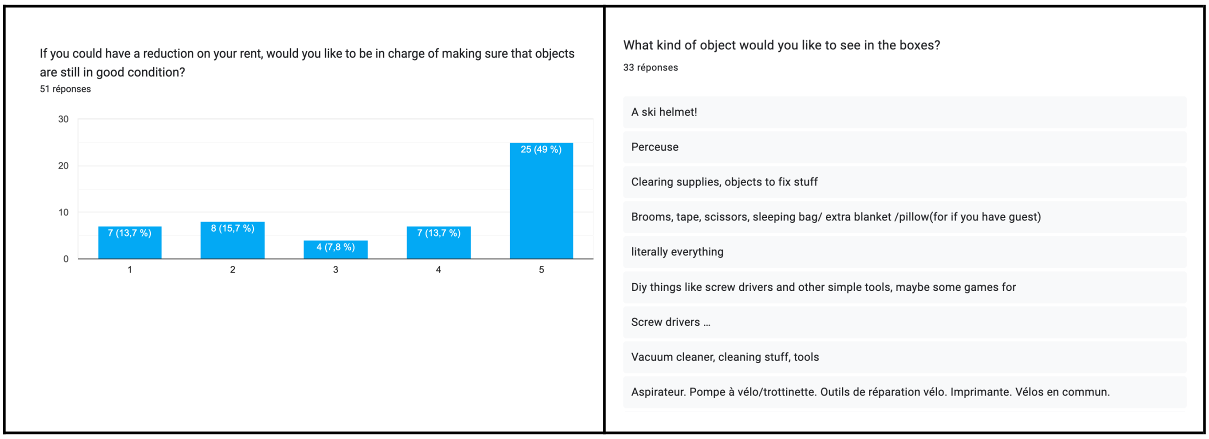


Figure B1 - Survey Results, 51 results

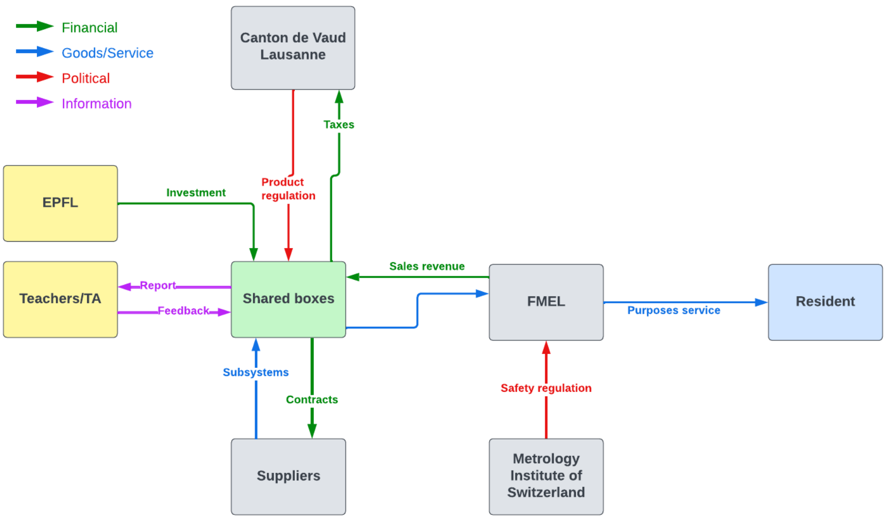


Figure B2 - Stakeholder analysis

Strengths

- Free service
- Low environmental impact
- Improves community spirit
- Promotes waste reduction

Weaknesses

- Community to create from scratch
- Time to establish
- Have not been prototyped yet

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Opportunities

- Residences trying to level up the standing/ the service
- Big interest at EPFL highlighted by a survey

Threats

- Relies on sharing
- Requires subsidies

Figure B3 - SWOT Analysis

C. Internal organization

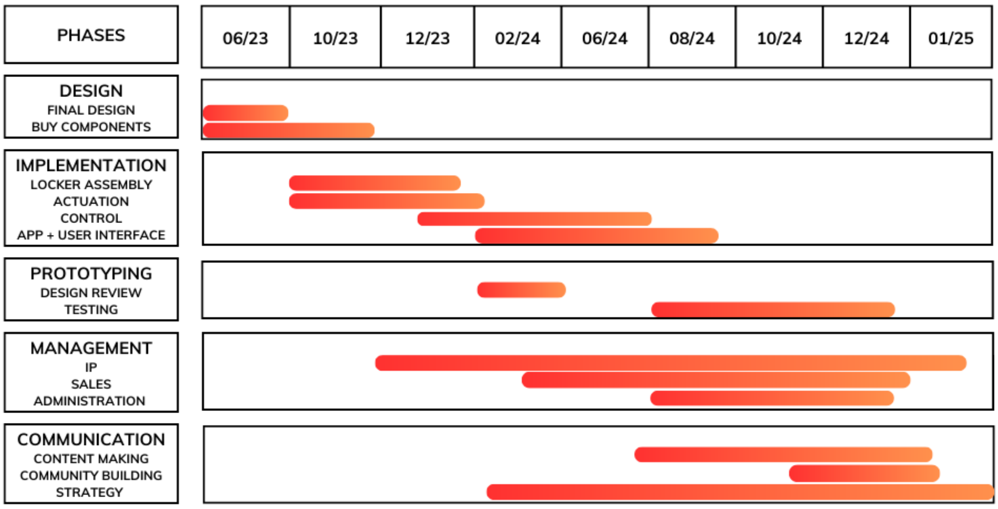

Figure C1 - Gantt Chart, schedule before launching

Figure C2 - Team Chart